EC 571: Digital VLSI Circuit Design

CMOS Logic

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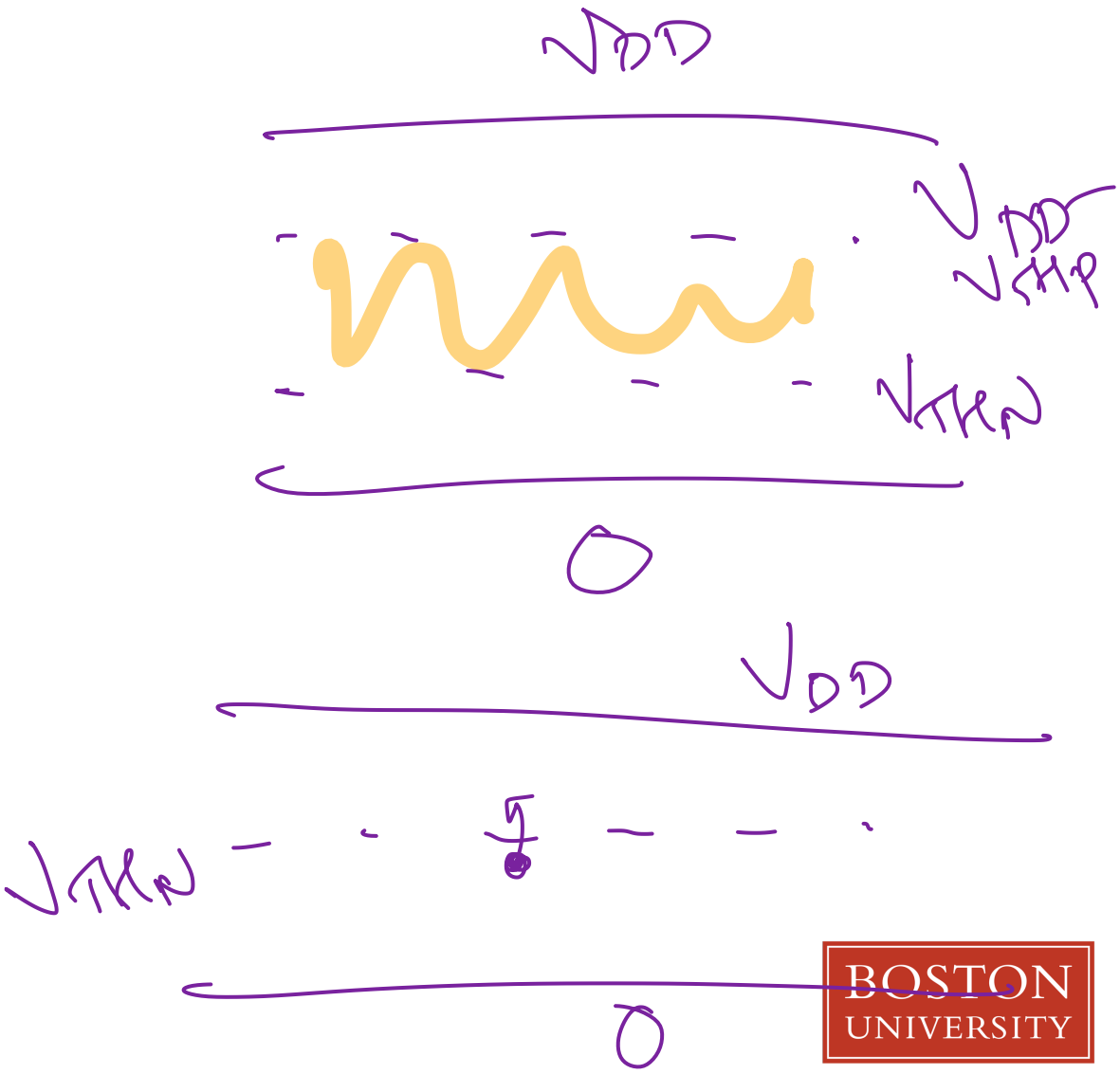
Acknowledgement - Some slides adapted from:

- J. Rabaey et al. Digital Integrated Circuits 2nd Ed.
- UC Berkeley EE141, UC Berkeley EE241, UC Berkeley EE251
- Stanford EE271

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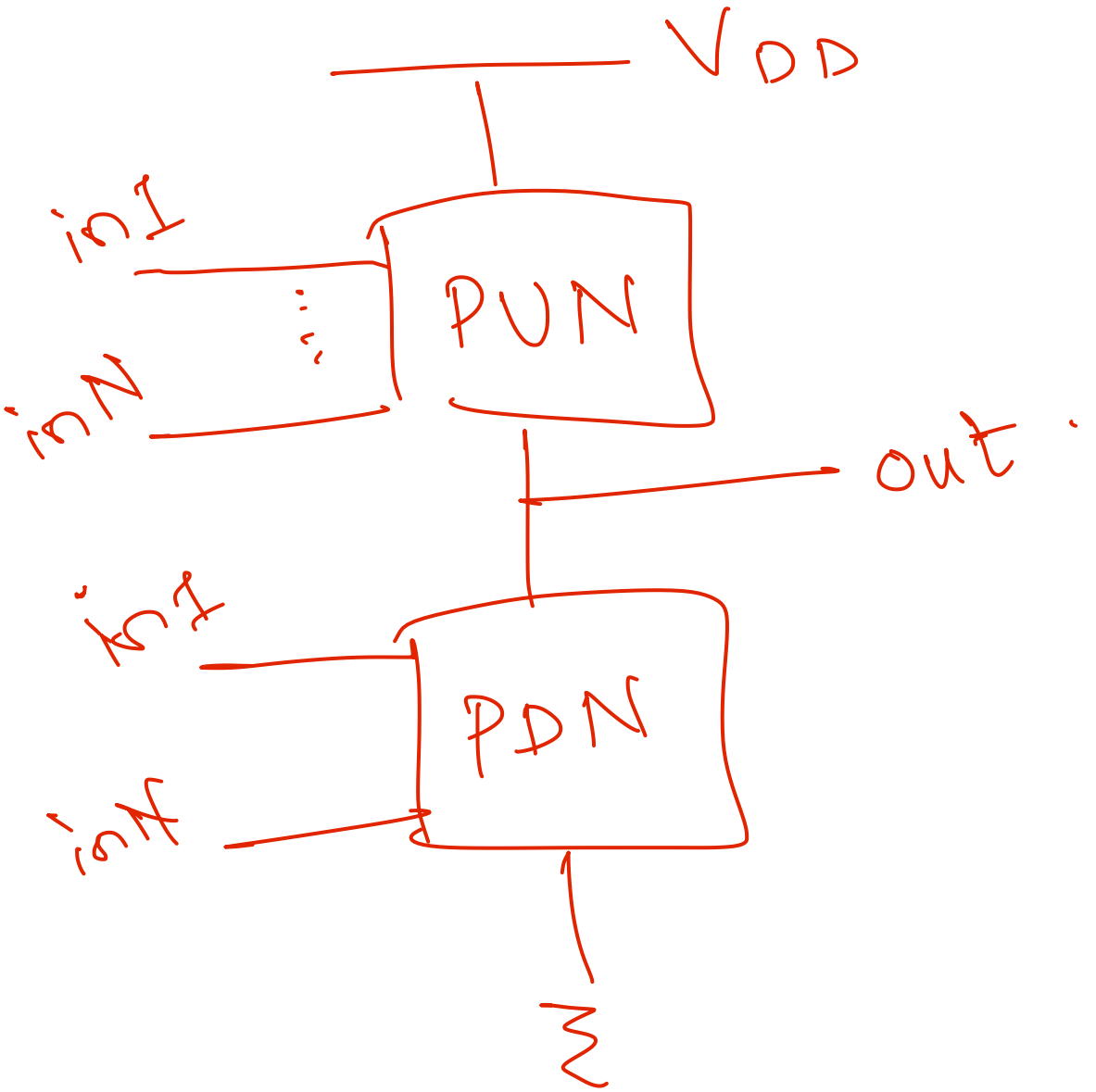
Last lecture

- Last lecture
  - Wires
- Today's lecture
  - CMOS logic
- Reading
  - Chapter 6

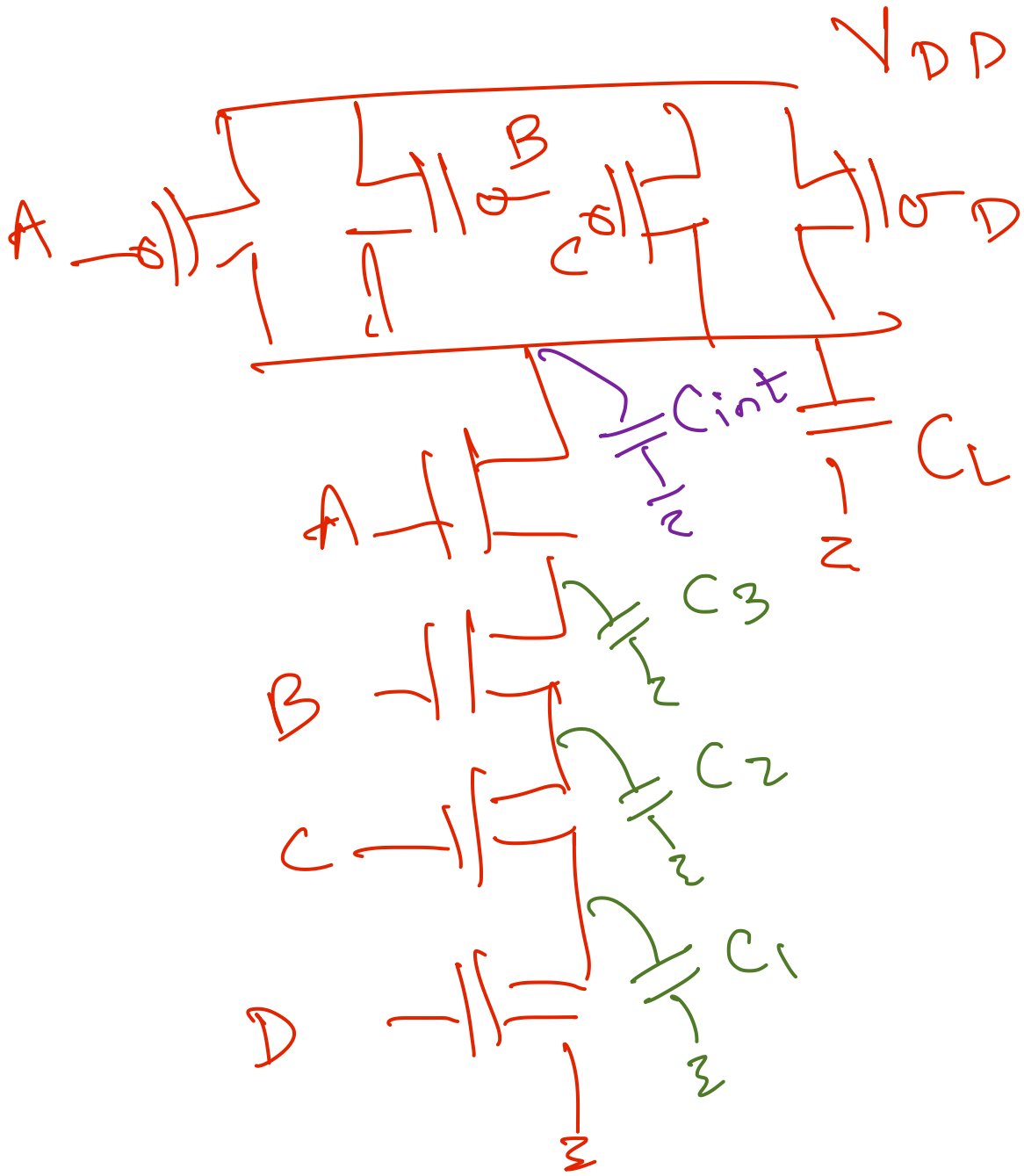


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Static Complementary CMOS Logic



NAND4 gate.

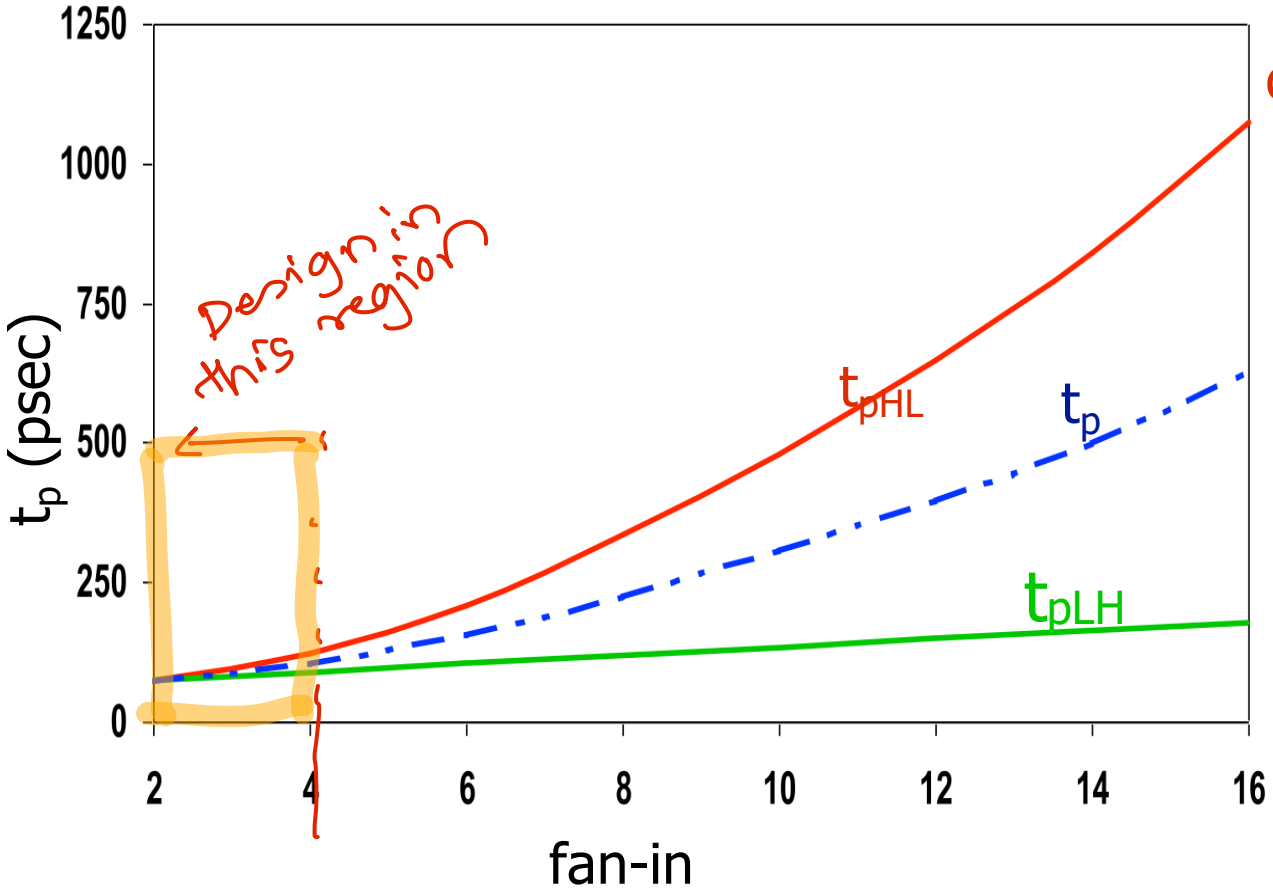


$$T_{HL} = (R_{n_A} + R_{n_B} + R_{n_C} + R_D) * (C_{int} + C_L) + (R_{n_B} + R_{n_C} + R_{n_D}) C_3 + (R_{n_C} + R_{n_D}) C_2 + R_{n_D} \cdot C_1$$

(1)

Propagation delay deteriorates rapidly as a function of fan in.

t_p as a Function of Fan-In



Gates with a fan-in greater than 4 should be avoided.

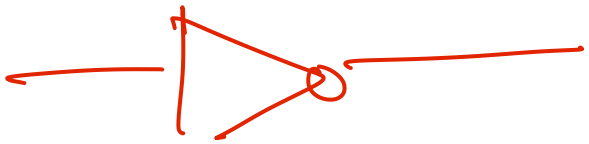
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Reducing parasitic delay

① Transistor Sizing

Xtor → W ↑ → R ↓ $\quad \tau \propto RC$

C ↑

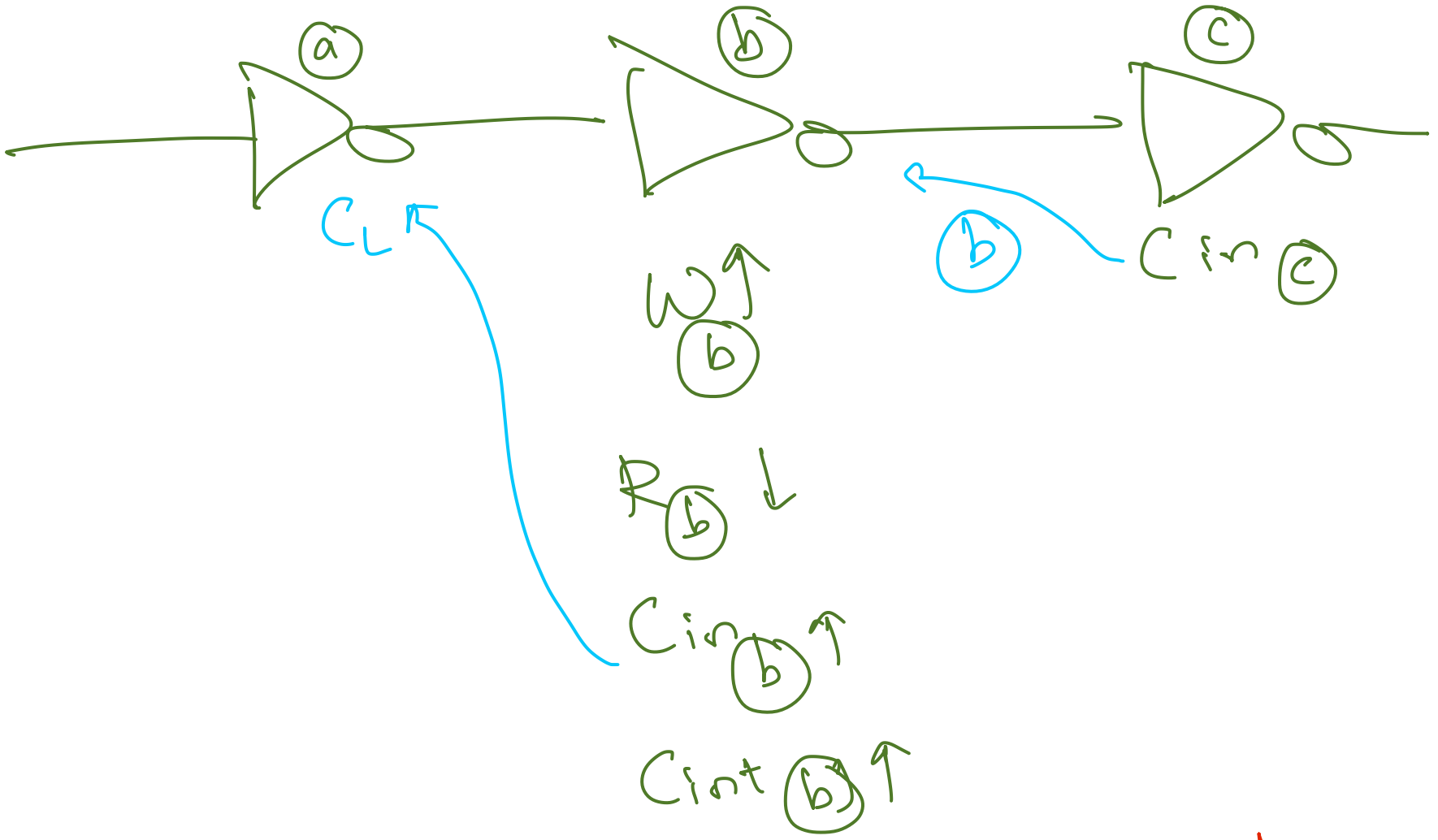


$\tau = R_n (C_{int} + C_L)$

$\tau = R_n \cdot C_{int} + R_n C_L$ (① $R_n \cdot C_{int}$, ② $R_n C_L$)

W_n ↑ ① −

② R ↓ C_L − ↓



- Dont think of gates in isolation
- Think of the entire combinational path. & have a balanced design.

(2) Progressive Transistor Sizing

From (1) on one of the previous slide

R_{n_D} appears 4 times

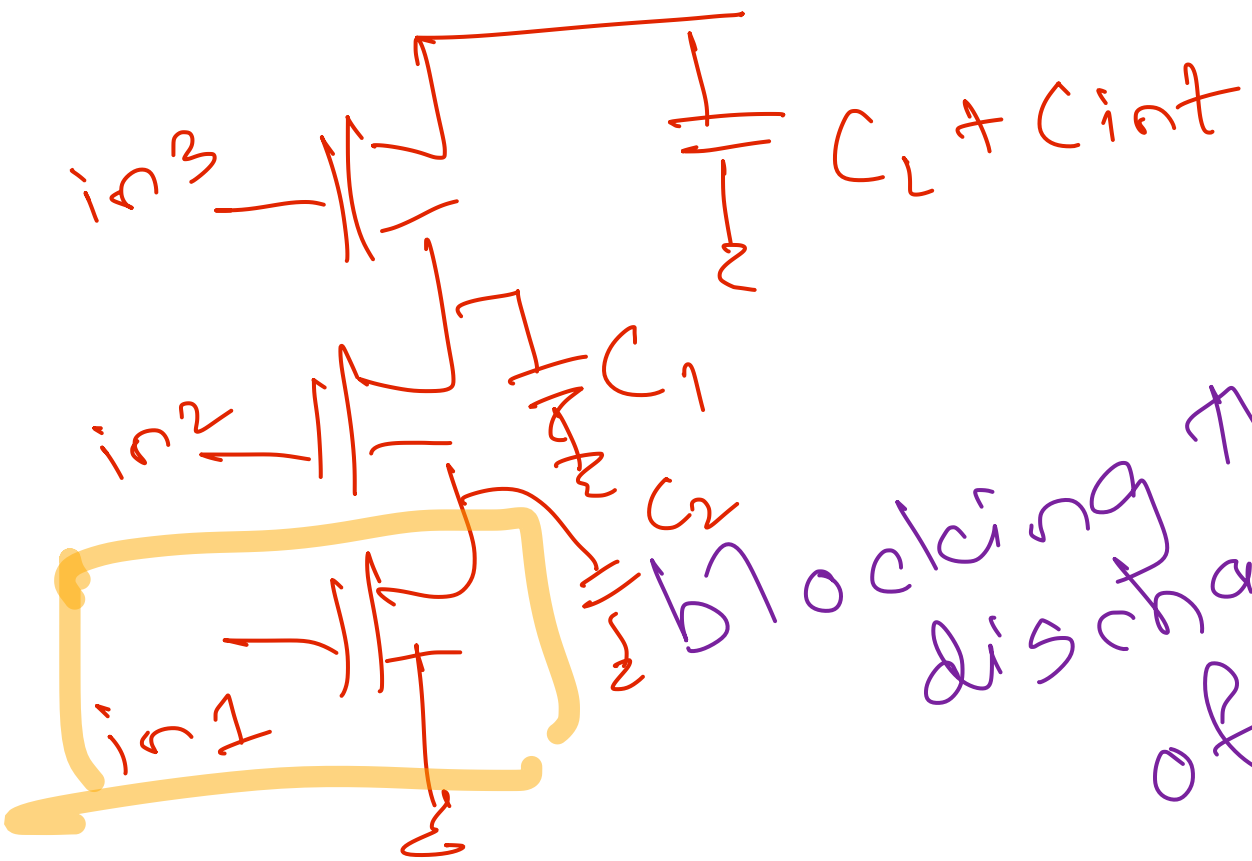
R_{n_C} " 3 "

R_{n_B} " 2 "

R_{n_A} " 1

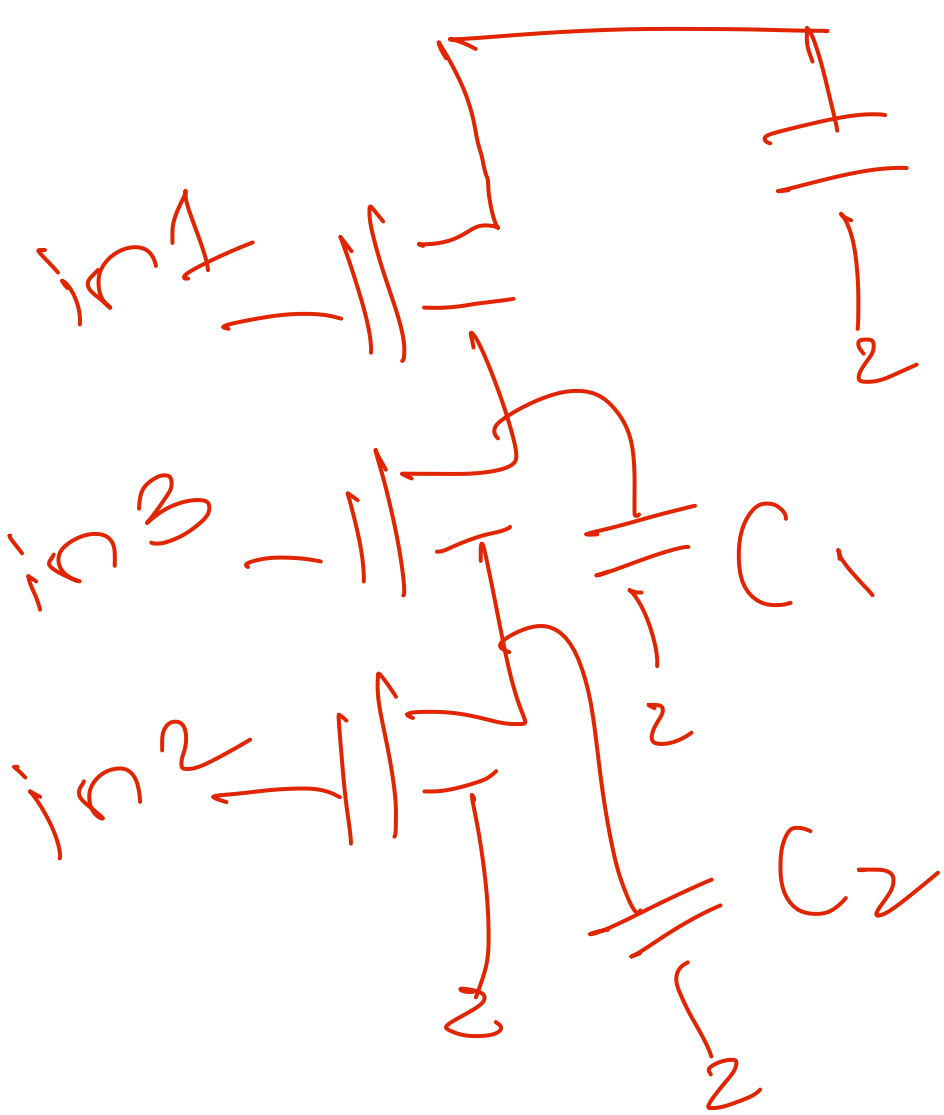
$$W_{n_D} > W_{n_C} > W_{n_B} > W_{n_A}$$

③ Input reordering



blocking the discharging of $C_L + C_{int}$ & C_1 & C_2

Assume in1 is the last to switch from 0 to 1



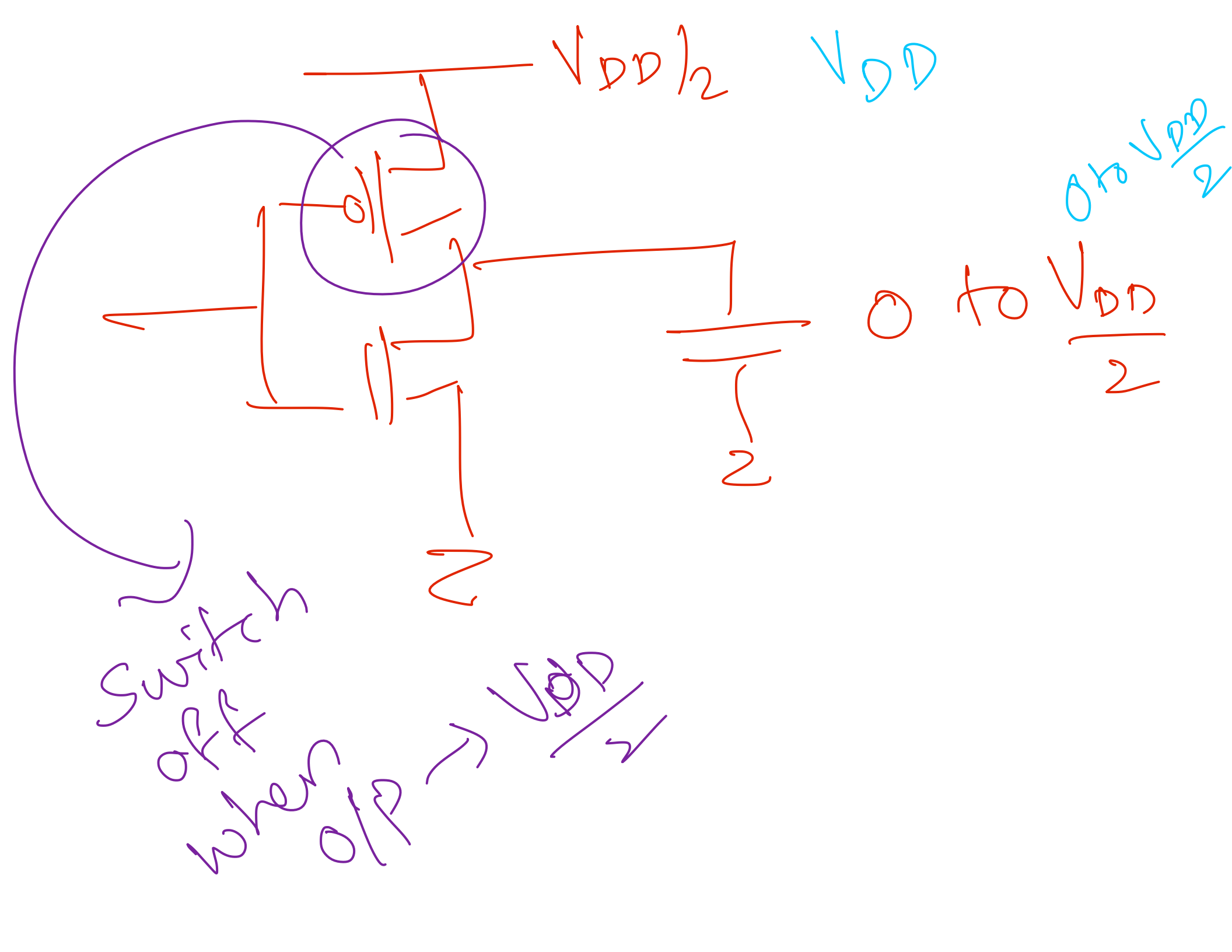
C_2 &
C_1 can start discharging before $in1$ changes from 0 to 1

④ Reduce voltage swing

$$t_{PHL} = 0.5 \frac{C_L V_{swing}}{I_{Sat_n}}$$

Instead of $0 \rightarrow V_{DD}$

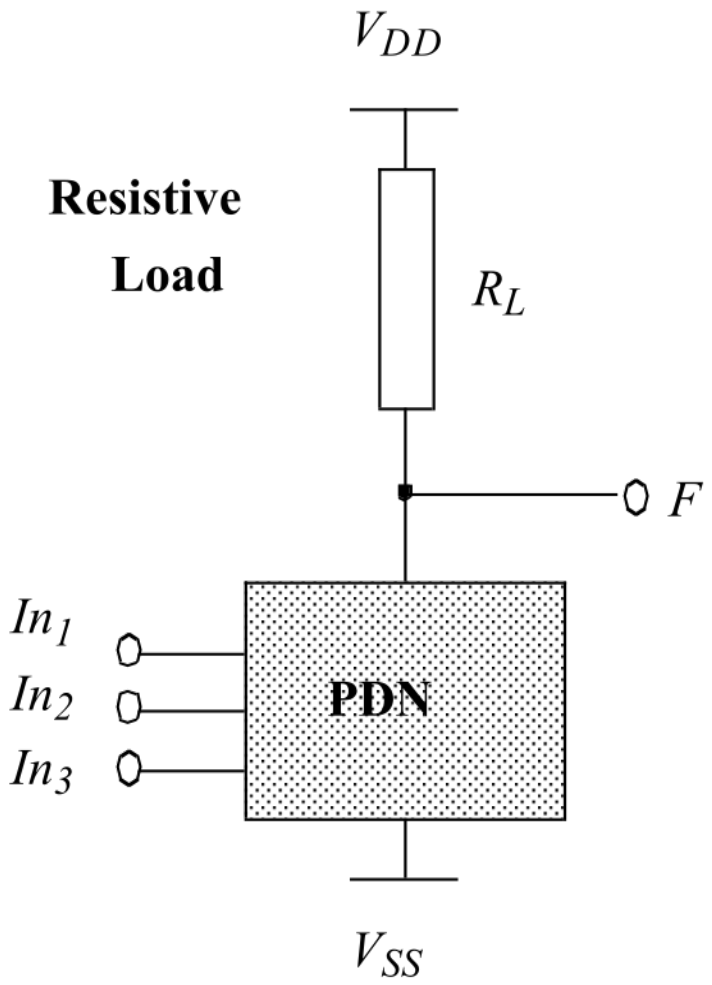
use $0 \rightarrow \frac{V_{DD}}{\alpha}$


$V_{DD}/2$
V_{DD}
0 to $\frac{V_{DD}}{2}$
0 to $\frac{V_{DD}}{2}$
Switch off when o/p → $\frac{V_{DD}}{2}$

Exam 1

Material upto this slide

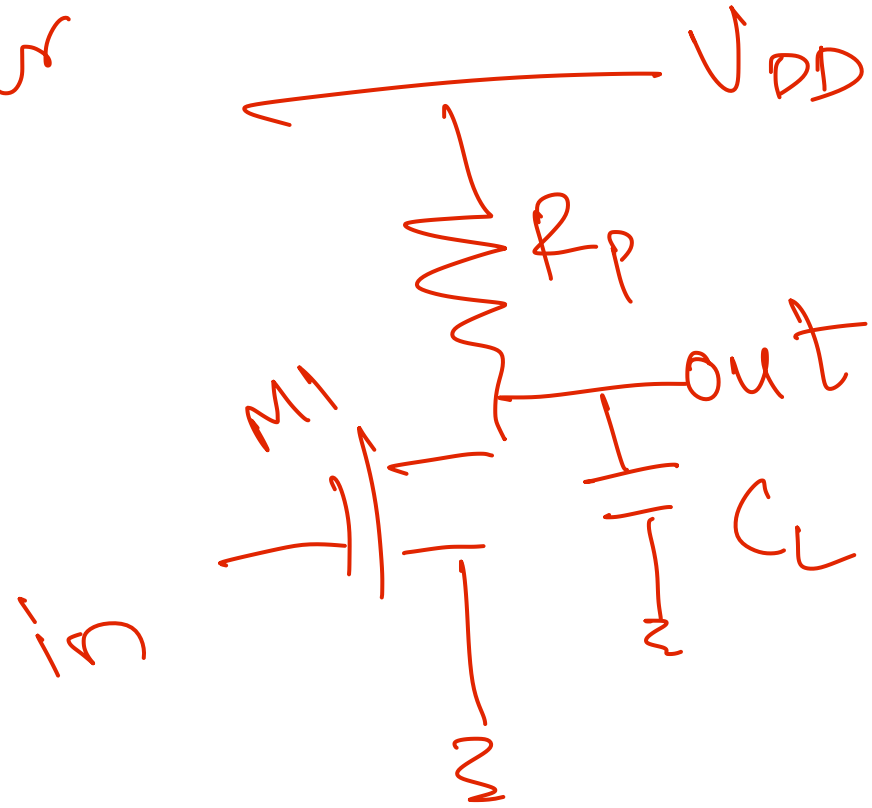
Ratioed Logic



- N transistors + Load
- $V_{OH} = V_{DD}$
- $V_{OL} = V_{DD} . R_N / (R_N + R_L)$
- Asymmetrical response
- Static power consumption

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Inverter



in = 0

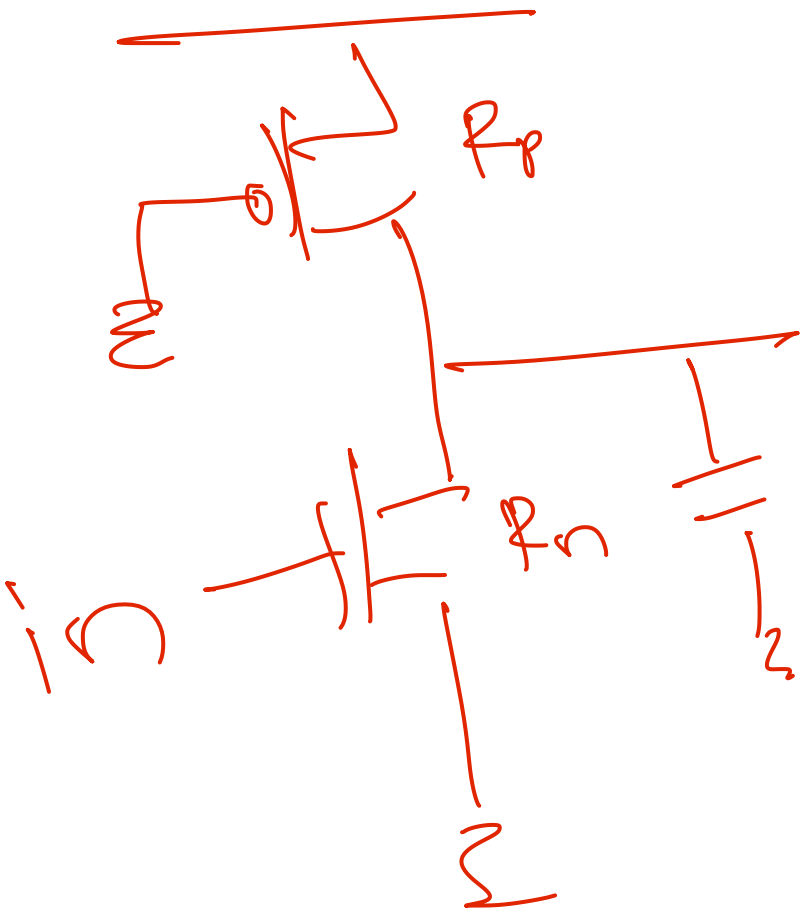
out = V_{DD}

in = 1

M1 is ON

out > 0V

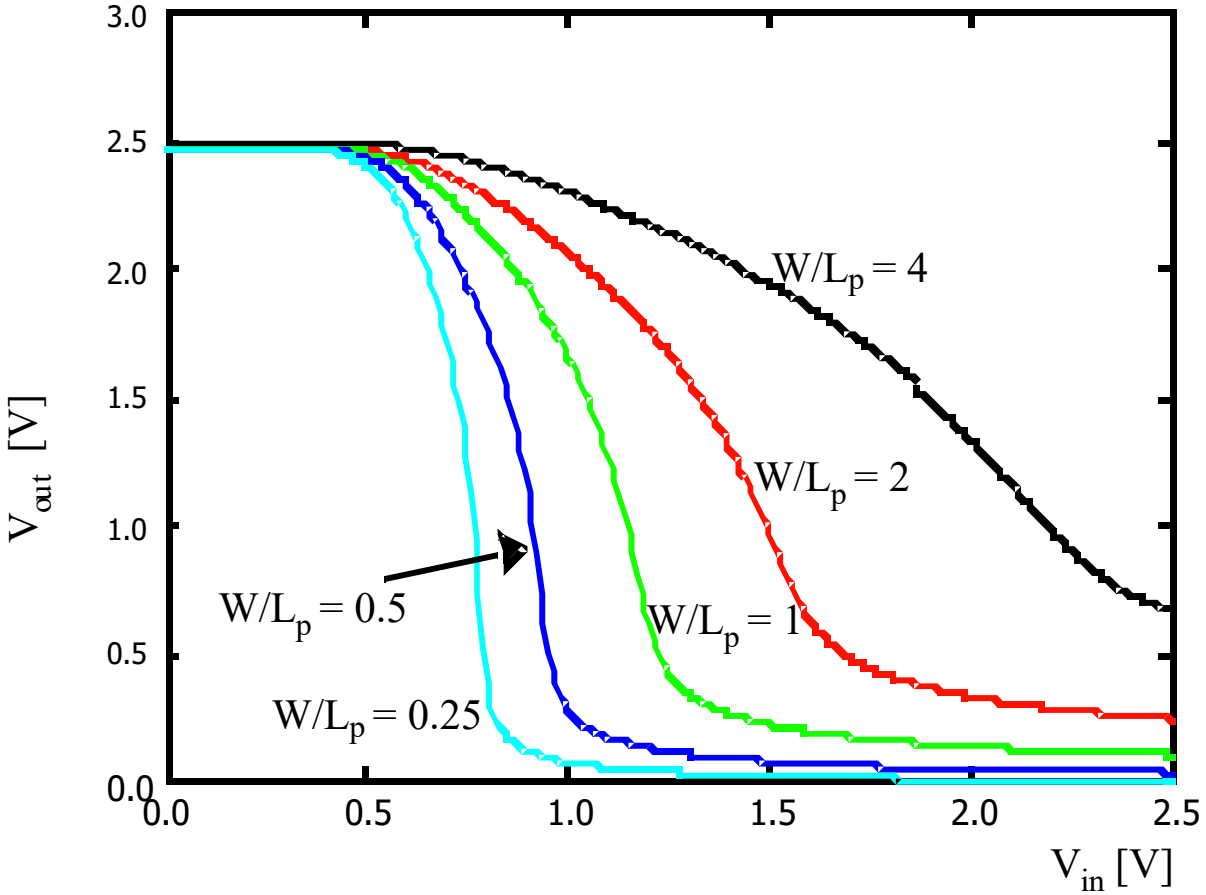
Value of out will depend on R_p & R_n



Behavior same as what described on the previous slide

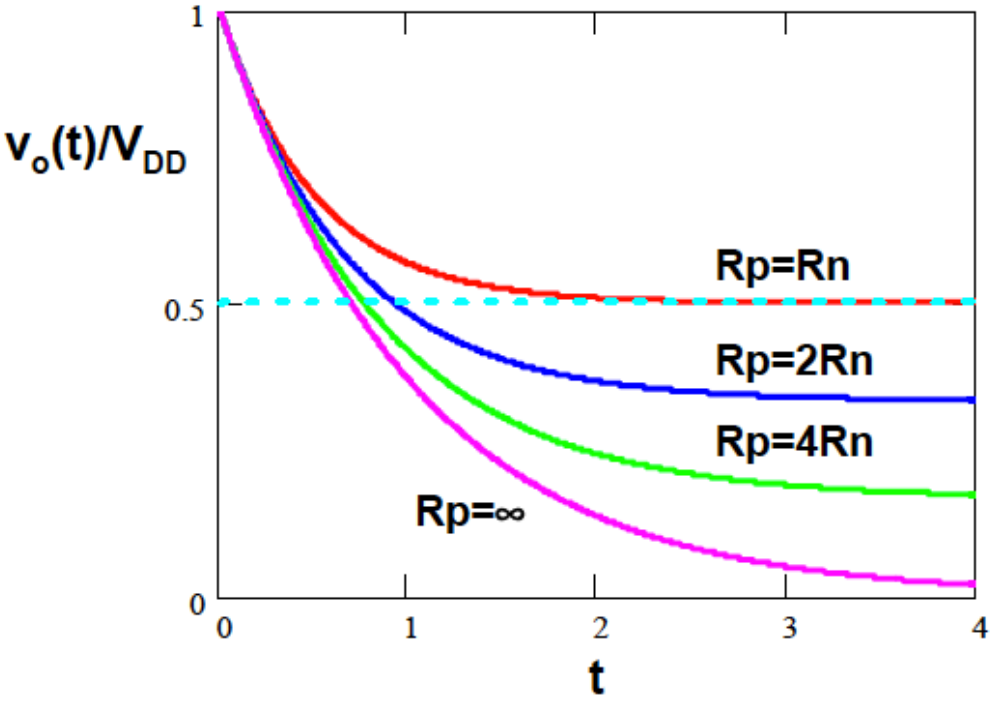
- No benefit in Xtor count for inverter
- For n-input gates
  'N' NMOS Xtors & PMOS Xtor

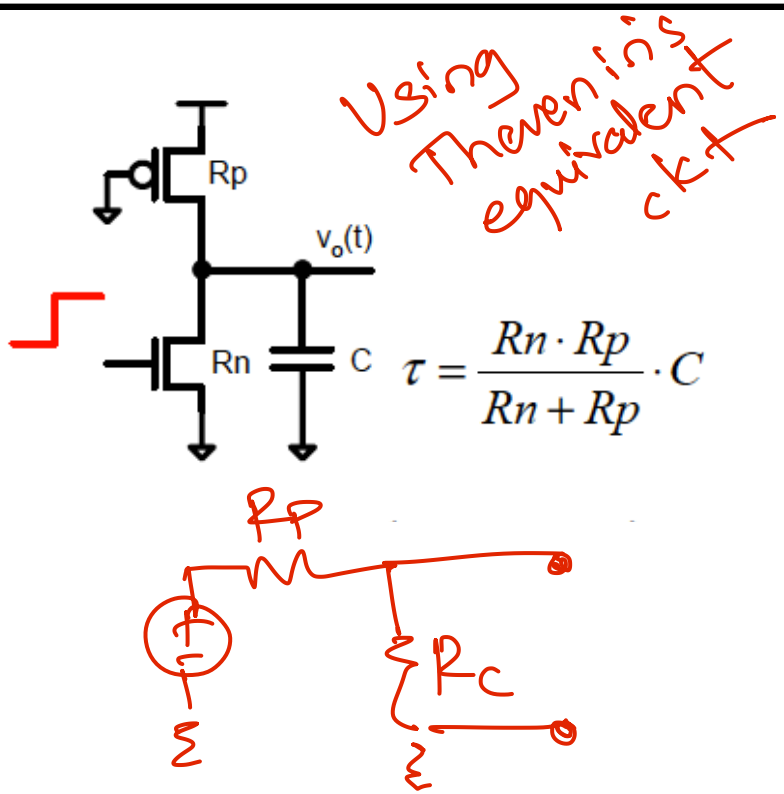
Pseudo-NMOS VTC



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Response on step input

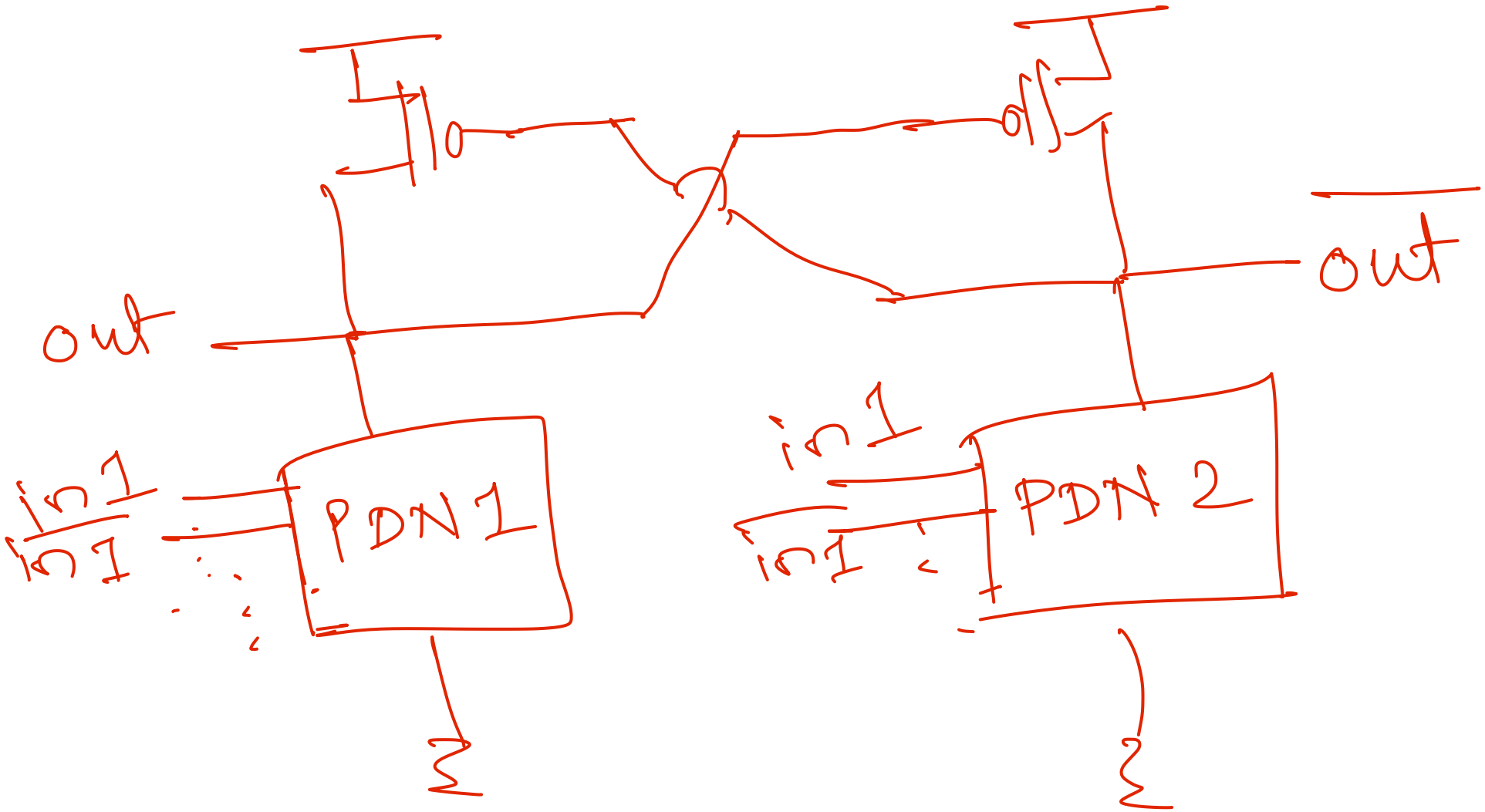




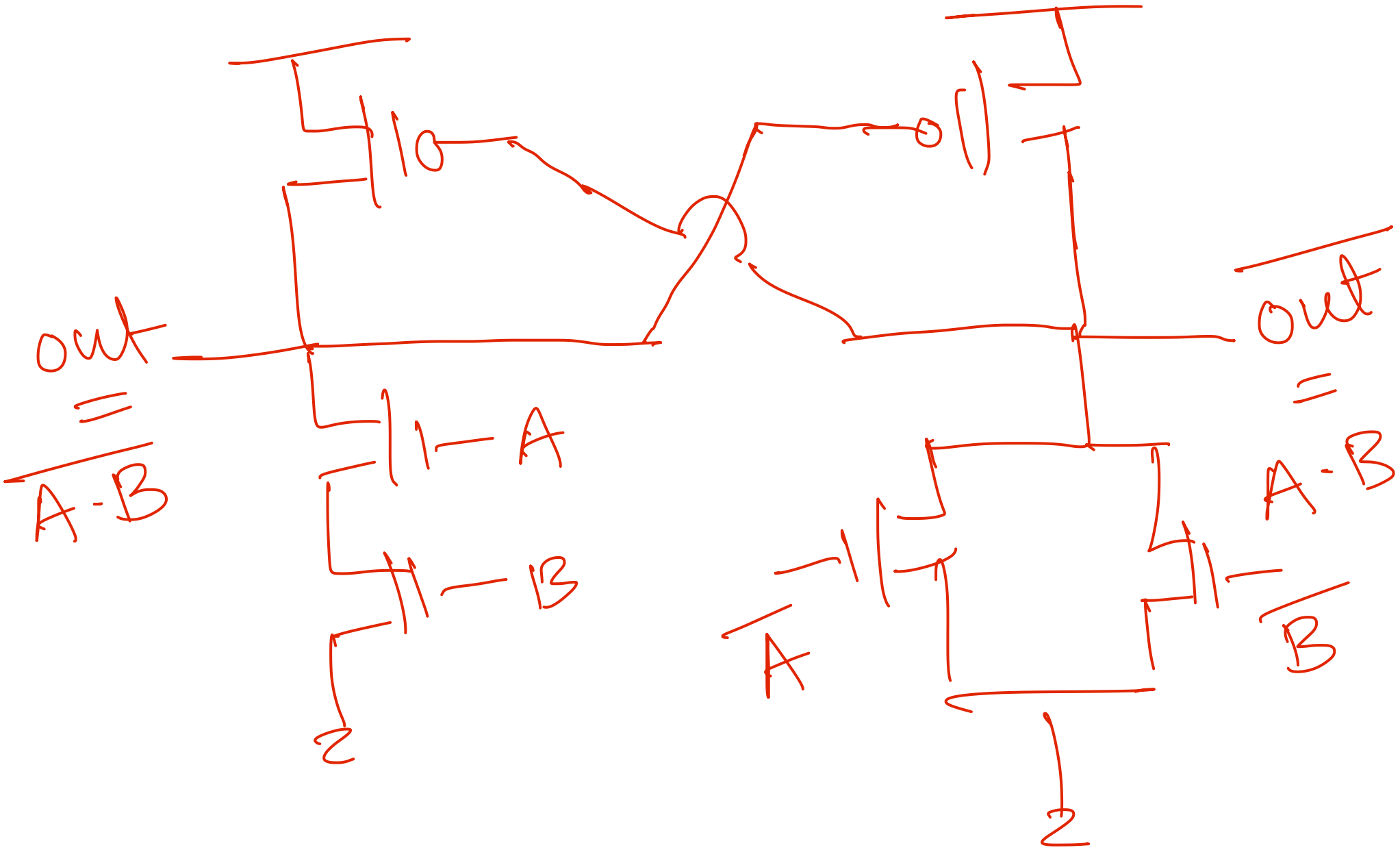
- Time constant is smaller, but it takes more time to complete 50% V_{DD} transient
  - Rp actually takes some current away from discharging C

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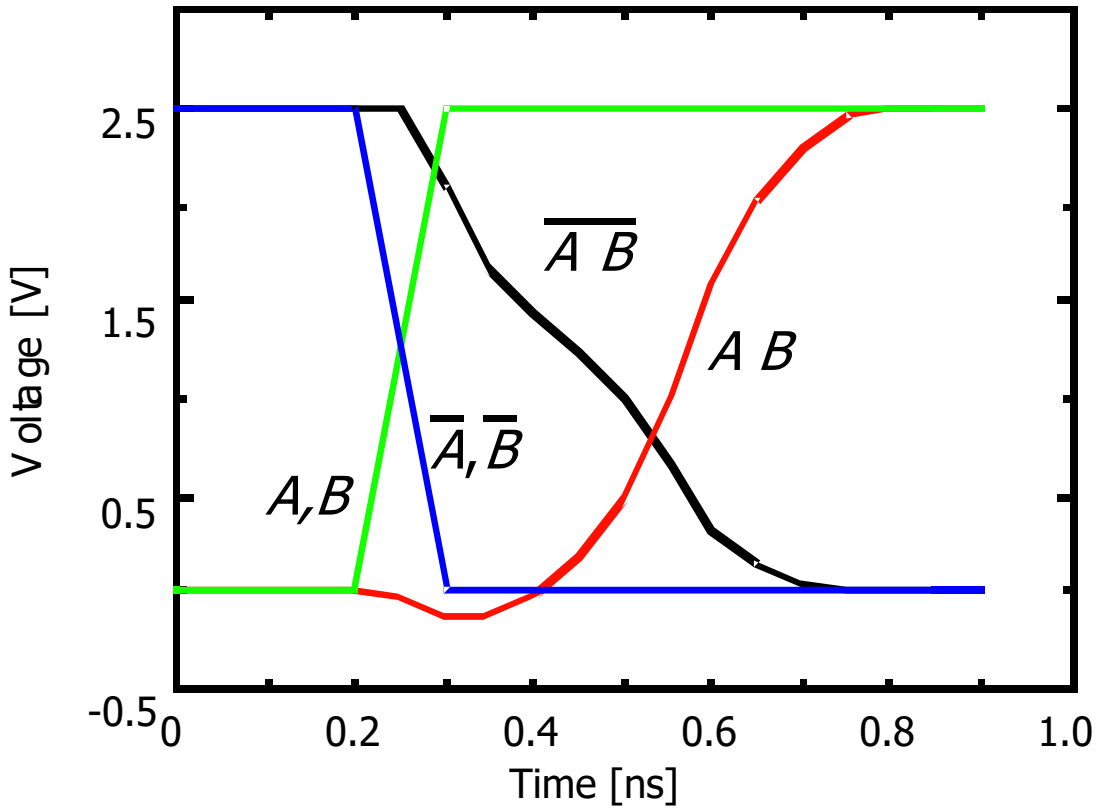
Differential Cascode Voltage Switch Logic



NAND2 / AND2 gate

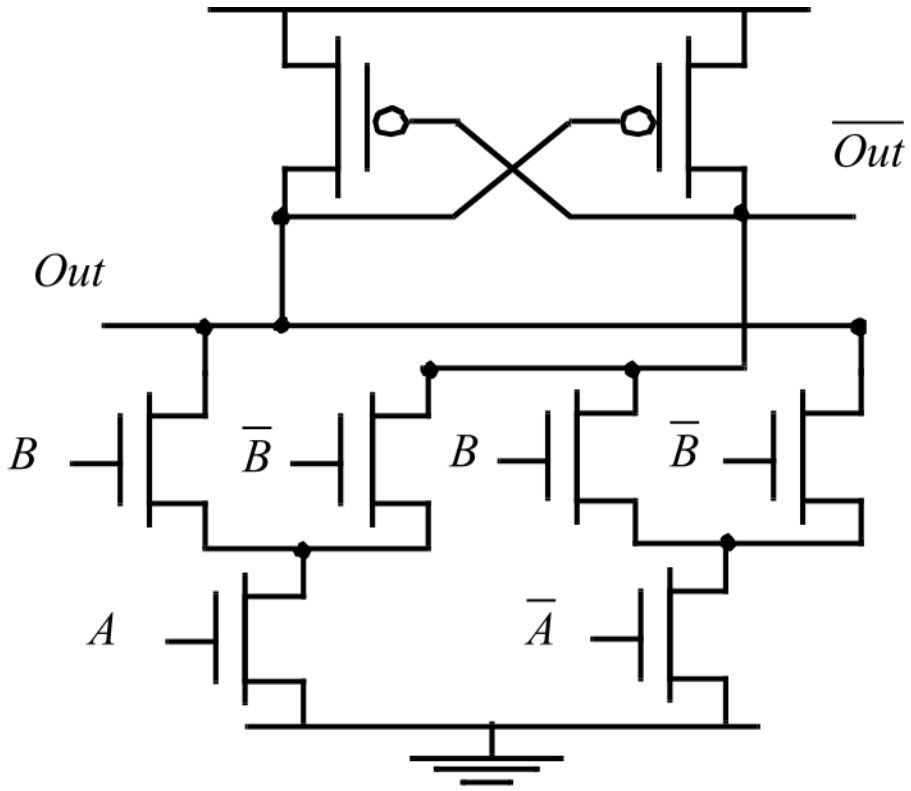


AND/NAND DCVSL Transient Response



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DCVSL Example



XOR-NXOR gate

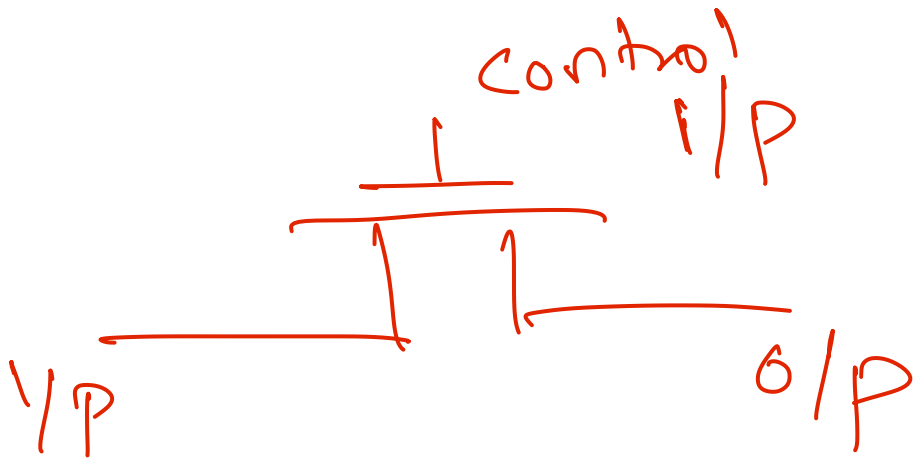
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Pass – transistor logic.

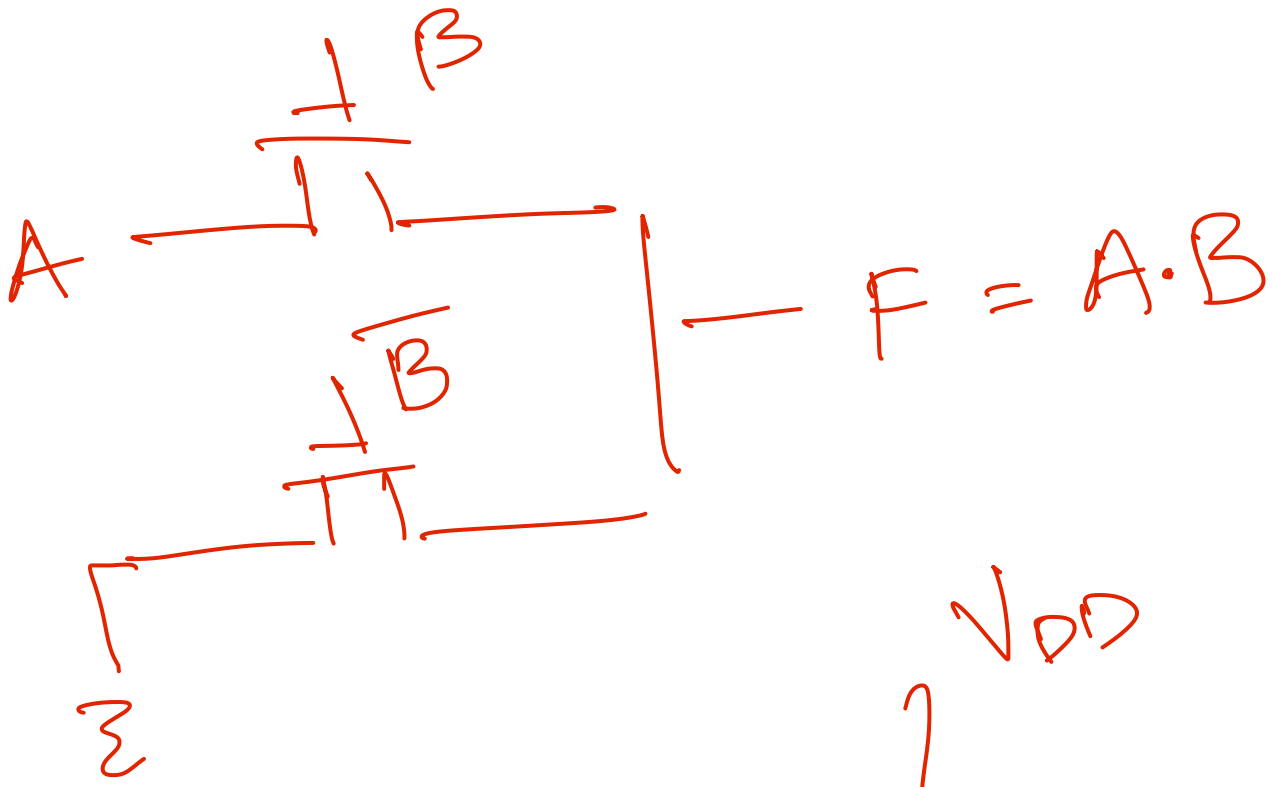
– Treat Xtors as switches

– Instead of O/P → V_{DD} or GND

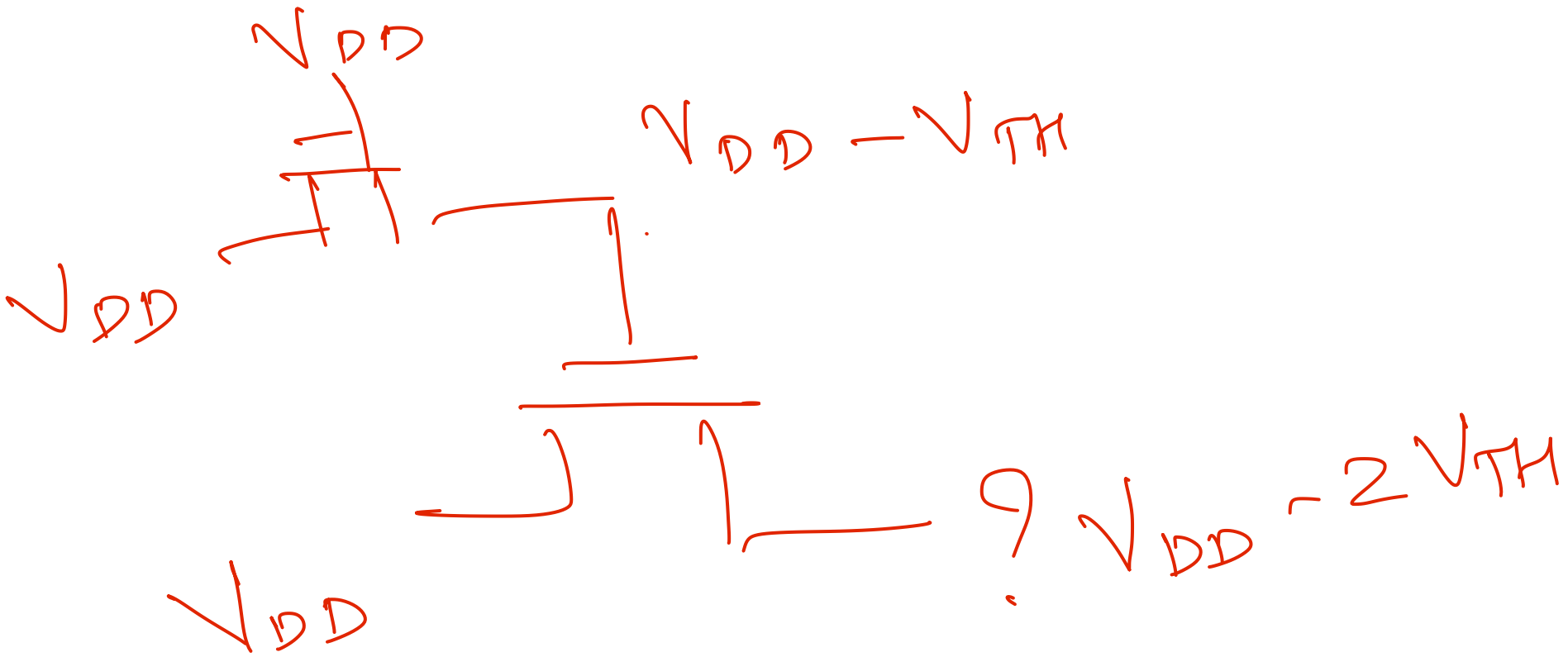
connect o/p to i/p



AND gate.






V_{DD}
V_{DD}
$V_{DD} - V_{TH}$
V_{DD}
?
$V_{DD} - 2V_{TH}$